Exercise 4.4

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Question 1.1

As in Exercises 4.2 and 4.3, for X = 1, 2, . . . , 100, simulate Y = 2 + 0.5X + , where ~ N(0, = 15). Set the seed to 1. This will be your "internal" training set. Report the regression coefficients, their standard errors, and the R^2.

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to bec
```

```
set.seed(1)
data_train <-  data.frame(
  X = 1:100) %>% mutate (Y = 2 + 0.5*X + rnorm(100, mean = 0, sd = 15))

lm_train <- lm(Y ~ X, data = data_train)
summary(lm_train)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data_train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -35.101  -9.088   0.233   8.777  34.462
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.97499    2.72846   1.457    0.148
## X            0.49323    0.04691  10.515   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.54 on 98 degrees of freedom
## Multiple R-squared:  0.5301, Adjusted R-squared:  0.5253
## F-statistic: 110.6 on 1 and 98 DF,  p-value: < 2.2e-16
```

Answer The estimated intercept was 3.9750 (SE = 2.7285), and the estimated slope was 0.4932 (SE = 0.0469). The fitted regression equation was:

$\hat{Y} = 3.9750 + 0.4932X$

The R-squared was 0.5301, indicating that the model explained approximately 53.0% of the variation in Y in the training set.

Question 1.2

Now, simulate an "external" test set. Simulate another 100 observations for X = 1, 2, . . . , 100, Y = 2 + 0.5X + , where ~ N(0, = 15). For this, set the seed to 2. This is the best-case scenario, as the model in the training set should fit the test set perfectly (except for sampling variation). In other words, the training set should show excellent discrimination, calibration and overall fit.

```r
set.seed(2)

data_test <- data.frame(X =1:100) %>% mutate(Y = 2 + 0.5 * X + rnorm(100, mean = 0, sd = 15))

head(data_test)
```

```
##   X          Y
## 1 1 -10.953718
## 2 2   5.772738
## 3 3  27.317680
## 4 4 -12.955635
## 5 5   3.296224
## 6 6   6.986304
```

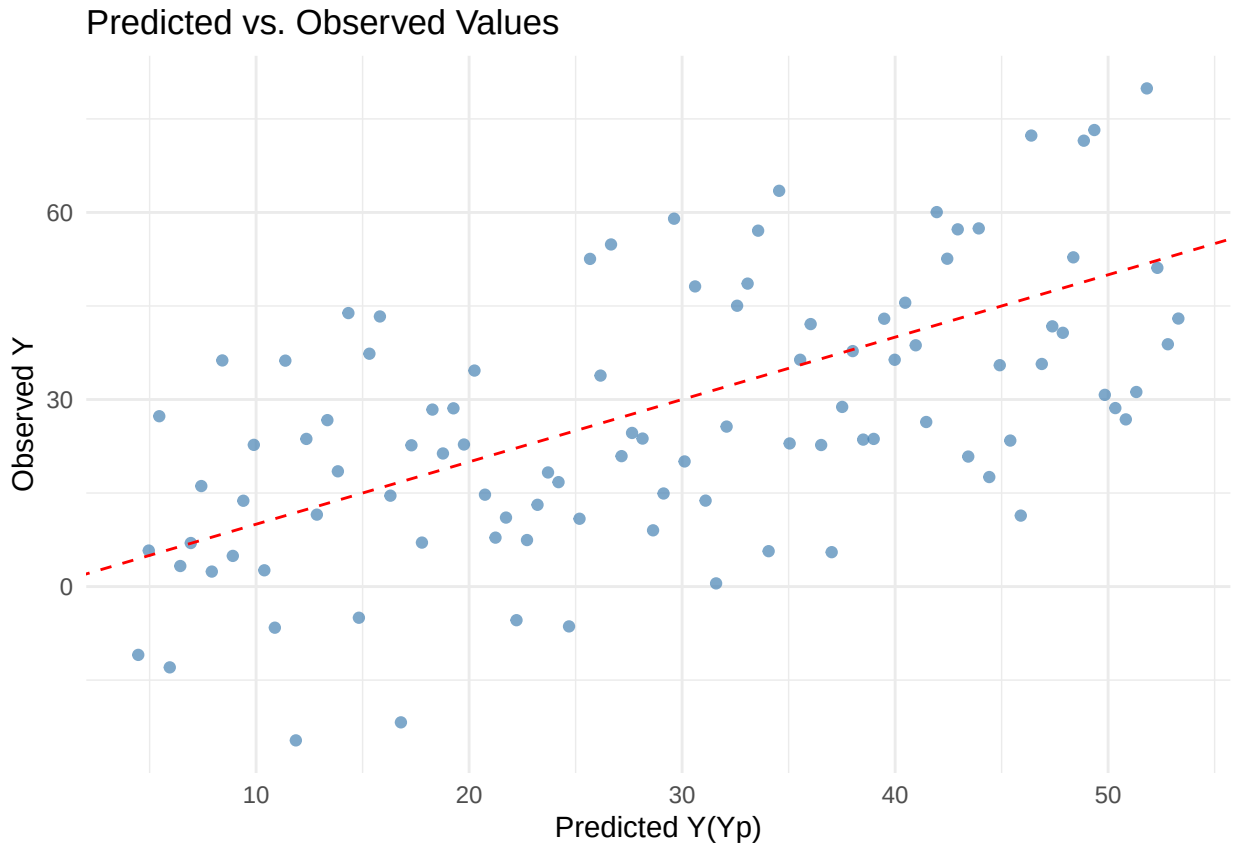
Answer I generated an external test set of 100 observations using the same data-generating model as the training set, with a different random seed (seed = 2).

Question 1.3

Apply the training model to the test set, generating predicted values Yp. Create a scatterplot of Yp (on the x-axis) versus Y (on the y-axis). Append a 45-degree line through the origin.

```
# Generate predicted values using the training model
data_test <- data_test %>%
  mutate(Yp = predict(lm_train, newdata = data_test))

# Scatterplot of predicted vs observed values
ggplot(data_test, aes(x = Yp, y = Y)) +
  geom_point(color = "steelblue", alpha = 0.7) +
  geom_abline(slope = 1, intercept = 0,
              color = "red", linetype = "dashed") +
  labs(title = "Predicted vs. Observed Values",
       x = "Predicted Y(Yp)",
       y = "Observed Y") +
  theme_minimal()
```



Answer The observed scatterplot compares the predicted values from the training model with the observed values in the external test set. The 45 degree refrence line represents perfect agreement between predicted and observed values.

Question 1.4

Assess discrimination. For example, one measure of discrimination is a non-parametric (e.g., Spearman Rank) correlation coefficient between Y and Yp.

```
# Assess discrimiation using Spearman rank correlation
cor(
  data_test$Y, data_test$Yp, method = "spearman"
)
```

```
## [1] 0.5770897
```

Answer The Spearman rank correlation between the observed predicted values was 0.578, indicating a moderate positive association. This suggests that the model has some ability to distinguish observations with higher versus lower Y values, although the ranking is not perfect.

Question 1.5

Assess calibration. One measure of calibration is a standardized mean difference of (Y-Yp). Hint: You can use the smd() function in the MBESS package.

```
# calcaulate the mean prediction error
mean(data_test$Y - data_test$Yp)
```

```
## [1] -2.093783
```

```
#Assess calibration using SMD
MBESS :: smd(data_test$Y, data_test$Yp)
```

```
## [1] -0.1159815
```

Answer The standardized mean difference was -0.116, indicating a relatively small difference between the observed and predicted means. The mean prediction error (Y − Yp) was -2.094, suggesting that the model slightly overpredicted Y on average. Overall, the results suggest relatively good calibration in terms of the average observed and predicted values.

Question 1.6

Assess overall fit. One measure of overall fit is R-squared. How does this R-squared statistic compare with the R-squared statistic from the training set? Hint: You can do this by hand (i.e., SSR/SST = 1 − SSE/SST) or use the R2() function in the caret package with form = "traditional".

```r
# Training R-squared
r2_train <- summary(lm_train)$r.squared

# Test R-squared
r2_test <- 1 -
  sum((data_test$Y - data_test$Yp)^2) /
  sum((data_test$Y - mean(data_test$Y))^2)

# compare training and test R-squared
r2_train
```

```
## [1] 0.5301334
```

```r
r2_test
```

```
## [1] 0.3145392
```

Answer The R-squared was 0.530 for the training set and 0.315 for the external test set. This indicates that the model explained approximately 53.0% of the variation in Y in the training set, compared with 31.5% in the test set.

The lower R-squared in the test set suggests weaker predictive performance on new data. Since both datasets were generated using the same model, the difference may be due to sampling variation.

Question 2

Now, simulate another test set. Simulate 100 observations for X = 1, 2, . . . , 100, Y = 10.5 + 0.5X + , where ~ N(0, = 15). Set the seed to 1. By changing the intercept, we have induced a consistent miscalibration.

Question 2.1

As before, create a scatterplot of Yp (on the x-axis) versus Y (on the y-axis). Append a 45-degree line through the origin.

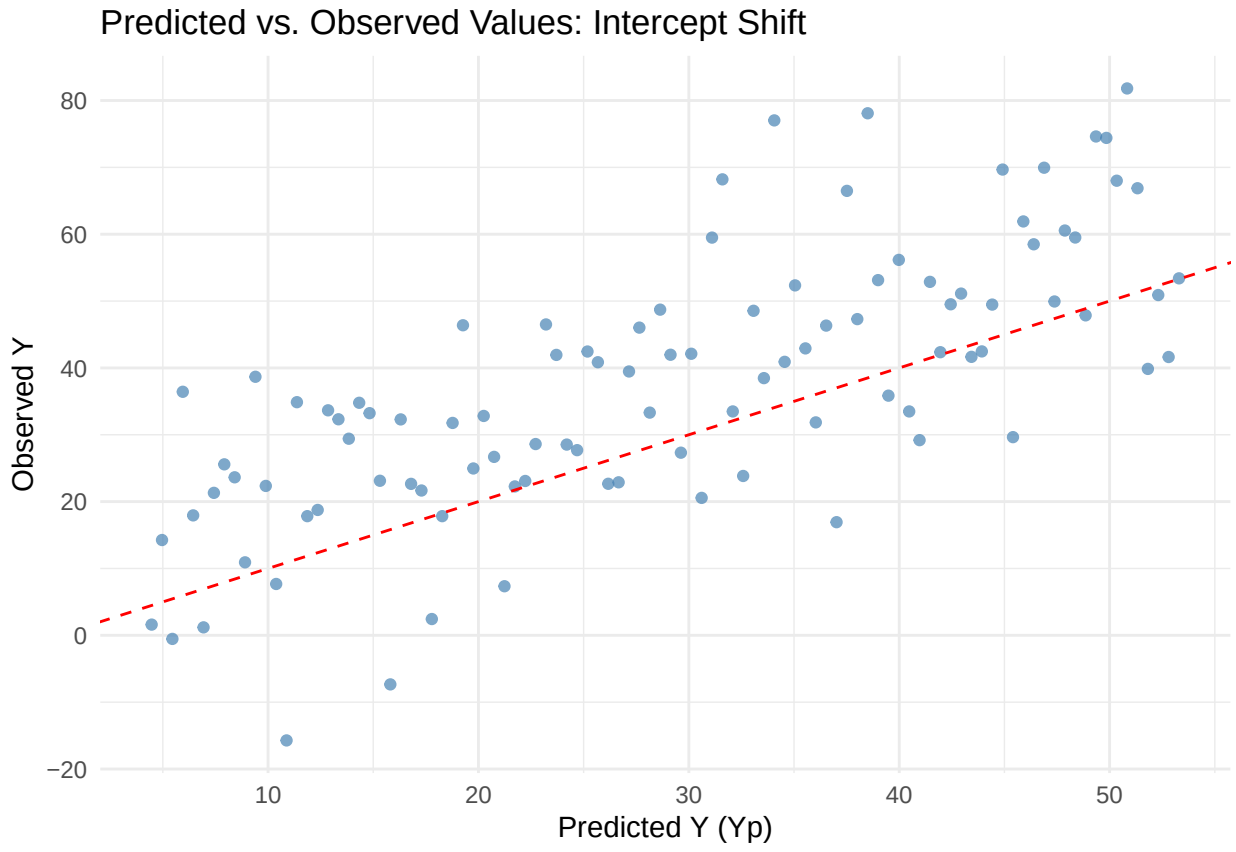
```r
# Simulate a new test set with a different intercept
set.seed(1)

data_test2 <- data.frame(
  X = 1:100
) %>%
  mutate(Y = 10.5 + 0.5 * X + rnorm(100, mean = 0, sd = 15))

# Apply the original training model
data_test2 <- data_test2 %>%
```

```r
  mutate(Yp = predict(lm_train, newdata = data_test2))

# Scatterplot
ggplot(data_test2, aes(x = Yp, y = Y)) +
  geom_point(color = "steelblue", alpha = 0.7) +
  geom_abline(slope = 1, intercept = 0,
              color = "red", linetype = "dashed") +
  labs(
    title = "Predicted vs. Observed Values: Intercept Shift",
    x = "Predicted Y (Yp)",
    y = "Observed Y"
  ) +
  theme_minimal()
```



Answer The scatterplot shows that many observed values fall above the 45-degree reference line, suggesting that the model systematically underpredicts Y. This is consistent with the increase in the intercept from 2 to 10.5 in the new test set.

Question 2.2

Assess discrimination.

```r
# Assess discrimination using Spearman rank correlation
cor(
  data_test2$Y,
  data_test2$Yp,
  method = "spearman"
)
```

```
## [1] 0.750351
```

Answer The Spearman rank correlation was 0.7504, indicating a strong positive association between the observed and predicted values. This suggests that the model maintains good discrimination despite the change in intercept. The higher correlation compared with Question 1 may reflect sampling variation rather than an improvement caused by the intercept shift.

Question 2.3

Assess calibration.

```r
# Mean prediction error
mean(data_test2$Y - data_test2$Yp)
```

```
## [1] 8.5
```

```r
# Standardized mean difference
MBESS::smd(
  data_test2$Y,
  data_test2$Yp
)
```

```
## [1] 0.4944685
```

Answer The standardized mean difference was 0.494, and the mean prediction error (Y − Yp) was 8.5. These results indicate that the model systematically underpredicts the observed values. Compared with Question 1, the larger standardized mean difference suggests poorer calibration, despite maintaining good discrimination.

Question 2.4

Assess overall fit.

```r
# Calculate test R-squared
r2_test2 <- 1 -
  sum((data_test2$Y - data_test2$Yp)^2) /
  sum((data_test2$Y - mean(data_test2$Y))^2)

# Compare with Question 1
r2_test
```

```
## [1] 0.3145392
```

```
r2_test2
```

```
## [1] 0.3411858
```

Answer The R-squared was 0.341 for the new test set, slightly higher than 0.315 in Question 1. Although the model showed systematic underprediction, its overall fit was slightly better in this simulation. This difference may be due to sampling variation because the two test sets were generated using different random seeds.

Question 3

Now, simulate another test set. Simulate 100 observations for X = 1, 2, . . . , 100, Y = X + , where ∼ N(0, = 15). Set the seed to 1. The model should continue to discriminate well, but perform more poorly otherwise.

Question 3.1

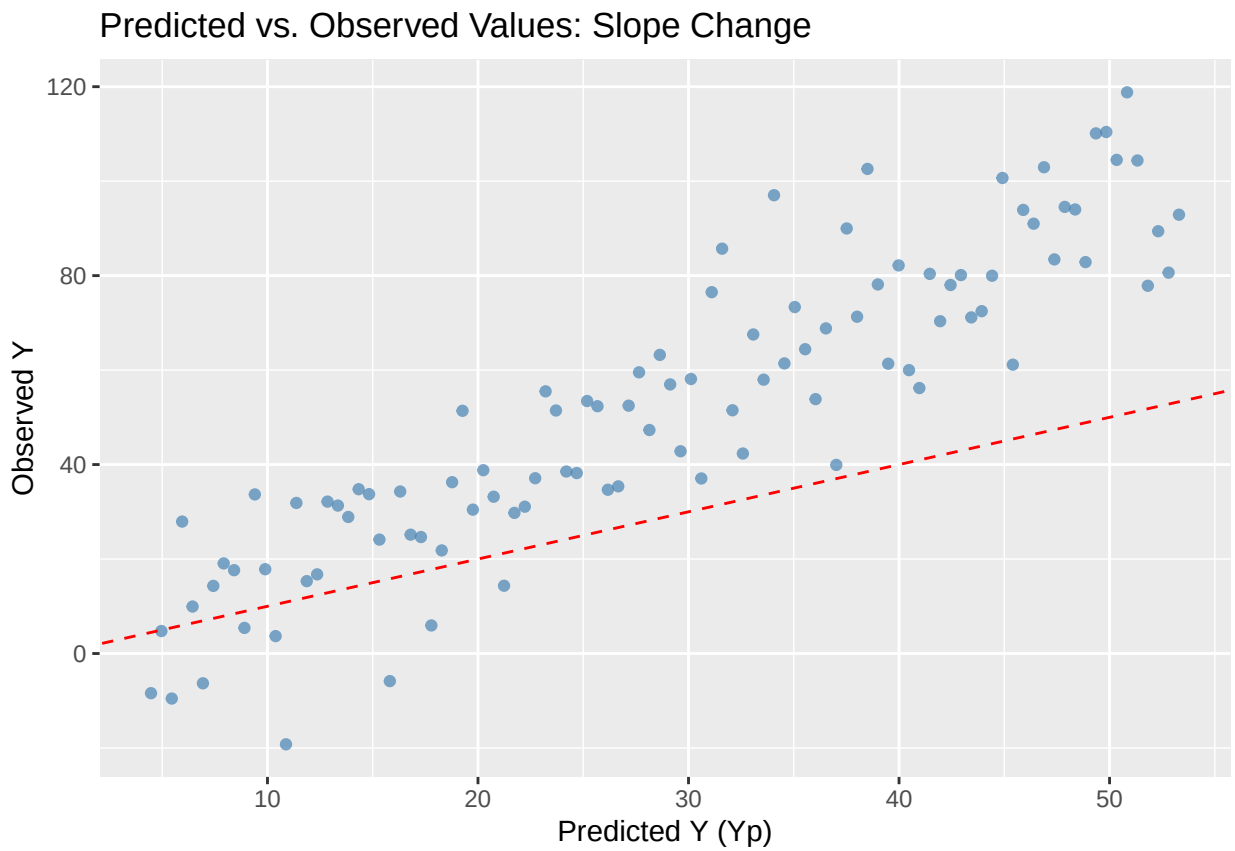
As before, create a scatterplot of Yp (on the x-axis) versus Y (on the y-axis). Append a 45-degree line through the origin.

```
#simulate a new test set
set.seed(1)

data_test3 <- data.frame(
  X = 1:100
) %>%
  mutate(Y = X + rnorm(100, mean = 0, sd = 15))

# Apply the original training model
data_test3 <- data_test3 %>%
  mutate(Yp = predict(lm_train, newdata = data_test3))

# Create scatterplot
ggplot(data_test3, aes(x = Yp, y = Y)) +
  geom_point(color = "steelblue", alpha = 0.7) +
  geom_abline(slope = 1, intercept = 0,
              color = "red", linetype = "dashed") +
  labs(
    title = "Predicted vs. Observed Values: Slope Change",
    x = "Predicted Y (Yp)",
    y = "Observed Y"
  )
```



Answer The scatterplot shows that most observed values fall above the 45-degree reference line, with the difference increasing as predicted values increase. This suggests that the model systematically underpredicts Y, especially at higher values, because the slope in the new test set is larger than that of the training model.

Question 3.2

Assess discrimination.

```
# Assess discrimination using Spearman rank correlation
cor(
  data_test3$Y,
  data_test3$Yp,
  method = "spearman"
)
```

```
## [1] 0.9234683
```

Answer The Spearman rank correlation was 0.9234, indicating very strong discrimination. Although the model systematically underpredicts Y, it is still able to distinguish observations with higher and lower outcome values.

Question 3.3

Assess calibration.

```
# Calculate mean prediction error
mean(data_test3$Y - data_test3$Yp)
```

```
## [1] 23.25
```

```
# Calculate standardized mean difference
MBESS::smd(
  data_test3$Y,
  data_test3$Yp
)
```

```
## [1] 0.9426951
```

Answer The standardized mean difference was 0.943, and the mean prediction error (Y − Yp) was 23.25. These results indicate substantial underprediction and poor calibration. The scatterplot also shows that underprediction increases at higher predicted values, consistent with the change in slope.

Question 3.4

Assess overall fit.

```
# Calculate test R-squared
r2_test3 <- 1 -
  sum((data_test3$Y - data_test3$Yp)^2) /
  sum((data_test3$Y - mean(data_test3$Y))^2)

# Compare test R-squared across Questions 1, 2, and 3
r2_test
```

```
## [1] 0.3145392
```

```
r2_test2
```

```
## [1] 0.3411858
```

```
r2_test3
```

```
## [1] 0.07301512
```

Answer The R-squared for the Question 3 test set was 0.073, which was substantially lower than 0.315 in Question 1 and 0.341 in Question 2. This indicates poor overall predictive fit. Despite strong discrimination, the model substantially underpredicted Y because the slope in the test set differed from that of the training model.

Question 3.5

Suppose that the investigator's goal is to identify patients who are likely to have very high values of Y based on X. These patients will receive an intervention, which is costly but effective. The investigator only has enough resources to give the intervention to 10% of patients. In plain English: is it sufficient for the model to show good discrimination, or does it also need to show good calibration and good overall fit?

Answer Good discrimination may be sufficient in this situation because the goal is to identify the 10% of patients most likely to have high Y values, rather than predict their exact outcomes. Even if the model systematically underpredicts Y, it can still rank patients correctly. Therefore, good calibration and overall fit are less important for this specific purpose, as long as the model reliably identifies the highest-risk patients.

Question 4

When an external dataset is unavailable then internal validation can still be performed. Internal validation, however, is more limited than external validation. First, the results can't be generalized to other populations. Second, the resulting inference isn't as strong or as extensive. In particular, instead of being able to test how well a specific model from the training set performs in the test set, all internal validation allows you to conclude pertains to the general properties of your model. Internal validation especially focuses on assessing the magnitude of overfitting – for example, by providing an estimate of R-squared that is unaffected by overfitting.

We will use a toy dataset to illustrate the structure of one form of internal validation.

Question 4.1

Fit a regression line to the entire dataset. Record the regression coefficients and R^2.

```
data_q4 <- data.frame(
  X = c(1, 2, 3, 4, 5),
  Y = c(2, 1.5, 2.5, 4.8, 7.2)
)

model_full <- lm(Y ~ X, data = data_q4)

summary(model_full)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data_q4)
##
## Residuals:
##     1     2     3     4     5
##  1.14 -0.73 -1.10 -0.17  0.86
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5100     1.1821  -0.431   0.6953
## X             1.3700     0.3564   3.844   0.0311 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.127 on 3 degrees of freedom
## Multiple R-squared:  0.8312, Adjusted R-squared:  0.775
## F-statistic: 14.77 on 1 and 3 DF,  p-value: 0.03107
```

Answer The estimated intercept was -0.510, and the slope was 1.370. The R-squared was 0.8312, indicating that the regression model explained approximately 83.1% of the variation in Y using all five observations.

Question 4.2

Delete the first observation (keep the remainder). Fit a regression line to the resulting dataset. Record the regression coefficients and R^2. Record the predicted value for the deleted observation (i.e., $X = 1$).

```r
# Remove the first observation
data_q4_1 <- data_q4[-1, ]

# Fit the regression model
model_q4_1 <- lm(Y ~ X, data = data_q4_1)

# Regression coefficients and R-squared
summary(model_q4_1)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data_q4_1)
##
## Residuals:
##     2     3     4     5
##  0.41 -0.53 -0.17  0.29
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2.7900     0.8710  -3.203   0.0852 .
## X             1.9400     0.2371   8.183   0.0146 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5301 on 2 degrees of freedom
```

```
## Multiple R-squared:  0.971,  Adjusted R-squared:  0.9565
## F-statistic: 66.97 on 1 and 2 DF,  p-value: 0.01461
```

```r
# Predict Y for the excluded observation (X = 1)
predict(model_q4_1, newdata = data_q4[1, , drop = FALSE])
```

```
##     1
## -0.85
```

Answer After excluding the first observation, the estimated intercept was -2.790, the slope was 1.940, and the R-squared was 0.971. The predicted Y value for the excluded observation (X = 1) was -0.850, compared with its actual value of 2.0.

Question 4.3

Delete the second observation (keep the remainder). Fit a regression line to the resulting dataset. Record the regression coefficients and R^2. Record the predicted value for the deleted observation (i.e., X = 2).

```r
# Remove the second observation
data_q4_2 <- data_q4[-2, ]

# Fit the regression model
model_q4_2 <- lm(Y ~ X, data = data_q4_2)

# Regression coefficients and R-squared
summary(model_q4_2)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data_q4_2)
##
## Residuals:
##       1       3       4       5
##  0.7229 -1.3086 -0.2743  0.8600
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.01143    1.49062   0.008   0.9946
## X            1.26571    0.41746   3.032   0.0937 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.235 on 2 degrees of freedom
## Multiple R-squared:  0.8213, Adjusted R-squared:  0.732
## F-statistic: 9.193 on 1 and 2 DF,  p-value: 0.09374
```

```r
# Predict Y for the excluded observation (X = 2)
predict(model_q4_2, newdata = data_q4[2, , drop = FALSE])
```

```
##        2
## 2.542857
```

Answer After excluding the second observation, the estimated intercept was 0.0114, the slope was 1.2657, and the R-squared was 0.8213. The predicted Y value for the excluded observation (X = 2) was 2.5429, compared with its actual value of 1.5.

Question 4.4

Repeat for the third observation.

```r
# Remove the third observation
data_q4_3 <- data_q4[-3, ]

# Fit the regression model
model_q4_3 <- lm(Y ~ X, data = data_q4_3)

# Regression coefficients and R-squared
summary(model_q4_3)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data_q4_3)
##
## Residuals:
##      1      2      4      5
##  0.865 -1.005 -0.445  0.585
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.235      1.150  -0.204   0.8569
## X              1.370      0.339   4.041   0.0561 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.072 on 2 degrees of freedom
## Multiple R-squared:  0.8909, Adjusted R-squared:  0.8363
## F-statistic: 16.33 on 1 and 2 DF,  p-value: 0.05613
```

```r
# Predict Y for the excluded observation (X = 3)
predict(model_q4_3, newdata = data_q4[3, , drop = FALSE])
```

```
##     3
## 3.875
```

Answer After excluding the third observation, the estimated intercept was -0.235, the slope was 1.370, and the R-squared was 0.8909. The predicted Y value for the excluded observation (X = 3) was 3.875, compared with its actual value of 2.5.

Question 4.5

Repeat for the fourth observation.

```r
# Remove the fourth observation
data_q4_4 <- data_q4[-4, ]

# Fit the regression model
model_q4_4 <- lm(Y ~ X, data = data_q4_4)

# Regression coefficients and R-squared
summary(model_q4_4)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data_q4_4)
##
## Residuals:
##       1       2       3       5
##  1.1400 -0.7543 -1.1486  0.7629
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5343     1.4492  -0.369   0.7477
## X             1.3943     0.4641   3.004   0.0952 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.373 on 2 degrees of freedom
## Multiple R-squared:  0.8186, Adjusted R-squared:  0.7279
## F-statistic: 9.025 on 1 and 2 DF,  p-value: 0.09524
```

```r
# Predict Y for the excluded observation (X = 4)
predict(model_q4_4, newdata = data_q4[4, , drop = FALSE])
```

```
##        4
## 5.042857
```

Answer After excluding the fourth observation, the estimated intercept was -0.5343, the slope was 1.3943, and the R-squared was 0.8186. The predicted Y value for the excluded observation (X = 4) was 5.0429, compared with its actual value of 4.8.

Question 4.6

Repeat for the fifth observation.

```
# Remove the fifth observation
data_q4_5 <- data_q4[-5, ]

# Fit the regression model
model_q4_5 <- lm(Y ~ X, data = data_q4_5)

# Regression coefficients and R-squared
summary(model_q4_5)
```

```
##
## Call:
## lm(formula = Y ~ X, data = data_q4_5)
##
## Residuals:
##     1     2     3     4
##  0.71 -0.73 -0.67  0.69
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.3500     1.2131   0.289    0.800
## X             0.9400     0.4429   2.122    0.168
##
## Residual standard error: 0.9905 on 2 degrees of freedom
## Multiple R-squared:  0.6925, Adjusted R-squared:  0.5387
## F-statistic: 4.504 on 1 and 2 DF,  p-value: 0.1678
```

```
# Predict Y for the excluded observation (X = 5)
predict(model_q4_5, newdata = data_q4[5, , drop = FALSE])
```

```
##    5
## 5.05
```

Answer After excluding the fifth observation, the estimated intercept was 0.350, the slope was 0.940, and the R-squared was 0.6925. The predicted Y value for the excluded observation (X = 5) was 5.050, compared with its actual value of 7.2.

Question 4.7

Now, fill in the table below. You should find that the regression parameters differ according to which observation is deleted. The R-squared statistics for each of these "training datasets" are potentially inflated by overfitting.

```r
# Combine the five leave-one-out regression models
models_q4 <- list(
  model_q4_1,
  model_q4_2,
  model_q4_3,
  model_q4_4,
  model_q4_5
)

# Create the summary table
results_q4 <- data.frame(
  Deleted = paste0("#", 1:5),
  Intercept = sapply(models_q4, function(m) coef(m)[1]),
  Slope = sapply(models_q4, function(m) coef(m)[2]),
  R_squared = sapply(models_q4, function(m) summary(m)$r.squared)
)

knitr::kable(results_q4, digits = 4)
```

Deleted	Intercept	Slope	R_squared
#1	-2.7900	1.9400	0.9710
#2	0.0114	1.2657	0.8213
#3	-0.2350	1.3700	0.8909
#4	-0.5343	1.3943	0.8186
#5	0.3500	0.9400	0.6925

Answer The regression coefficients and R-squared values varied depending on which observation was excluded. The R-squared values ranged from 0.6925 to 0.9710, suggesting that the fitted models were sensitive to individual observations. These training R-squared values may overestimate predictive performance on unseen data.

Question 5

Now, create a data frame by (1) copying each of the predicted values from Question 4; and (2) copying the actual observed values of Y.

Question 5.1

Fit a simple linear regression to this data (i.e., Y ~ Yp). Record the regression coefficients and R^2.

```r
# Create a data frame with actual and predicted Y values
data_q5 <- data.frame(
  Y = data_q4$Y,
  Yp = c(
```

```
    predict(model_q4_1, newdata = data_q4[1, , drop = FALSE]),
    predict(model_q4_2, newdata = data_q4[2, , drop = FALSE]),
    predict(model_q4_3, newdata = data_q4[3, , drop = FALSE]),
    predict(model_q4_4, newdata = data_q4[4, , drop = FALSE]),
    predict(model_q4_5, newdata = data_q4[5, , drop = FALSE])
  )
)

# View the data
data_q5
```

```
##     Y        Yp
## 1 2.0 -0.850000
## 2 1.5  2.542857
## 3 2.5  3.875000
## 4 4.8  5.042857
## 5 7.2  5.050000
```

```
# Fit the regression model
model_q5 <- lm(Y ~ Yp, data = data_q5)

# Report coefficients and R-squared
summary(model_q5)
```

```
##
## Call:
## lm(formula = Y ~ Yp, data = data_q5)
##
## Residuals:
##        1        2        3        4        5
##  1.04350 -1.70881 -1.59314 -0.06841  2.32685
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.5208     1.5564   0.977    0.401
## Yp            0.6638     0.4069   1.631    0.201
##
## Residual standard error: 1.997 on 3 degrees of freedom
## Multiple R-squared:  0.4701, Adjusted R-squared:  0.2934
## F-statistic: 2.661 on 1 and 3 DF,  p-value: 0.2013
```

Answer The estimated intercept was 1.5208, and the slope was 0.6638. The R-squared was 0.4701, indicating a weaker linear relationship between the observed and leave-one-out predicted values than the fit suggested by the original model.

Question 5.2

Is the value of R^2 smaller than the values in the above table? In plain English, what should happen when the original dataset is overfit?

Answer Yes. The R-squared of 0.4701 was lower than all the R-squared values in Question 4, which ranged from 0.6925 to 0.9710.

When a model is overfit, it performs well on the data used to build it but does not predict new observations as well. As a result, its predictive performance is usually worse on data that were not used to fit the model.